

DTF via Robust Scaling: Replication Code for Simulation 1

Blinded

2026-08-06

Initialization

```
library(robustDIF)
library(dplyr)
library(tidyr)
library(ggplot2)
library(see)

check_archive_robustDIF <- function(version = "0.2.0.20260708",
                                     archive.name = "review-archive") {
  installed_version <- as.character(utils::packageVersion("robustDIF"))
  if (installed_version != version) {
    stop(
      "This report expects robustDIF ", version,
      ", but found robustDIF ", installed_version,
      ". Install robustDIF from the archive branch.",
      call. = FALSE)
  }

  desc <- utils::packageDescription("robustDIF")
  installed_archive <- desc[["Config/robustDIF/archive-name"]]
  if (!identical(installed_archive, archive.name)) {
    stop(
      "This report expects robustDIF archive '", archive.name,
      "'", but found archive label '", installed_archive,
      "'. Install robustDIF from the archive branch.",
      call. = FALSE)
  }
}
```

```

}
invisible(installed_version)
}

check_archive_robustDIF()

find_results_dir <- function(candidates, pattern, min_files) {
  candidates <- unique(candidates)
  n_files <- vapply(
    candidates,
    function(path) {
      if (!dir.exists(path)) return(0L)
      length(list.files(path, pattern = pattern, full.names = TRUE))
    },
    integer(1))

  if (!any(n_files >= min_files)) {
    stop(
      "Could not find the Simulation 1 result files. Expected at least ",
      min_files, " files matching ", pattern, " in one of: ",
      paste(candidates, collapse = ", "),
      call. = FALSE)
  }

  normalizePath(candidates[which(n_files >= min_files)[1]], mustWork = TRUE)
}

sim1_pattern <- "^sim1_n[0-9]_ndif[0-9]+[.]rds$"
results_dir <- find_results_dir(
  c(
    "../results/sim1",
    "results/sim1",
    "../../paper/results/sim1",
    "../../../paper/results/sim1",
    "paper/results/sim1"),
  pattern = sim1_pattern,
  min_files = 33L)

method_levels <- c("LRT", "RDIF", "True")

method_cols <- c(
  "LRT" = "#254A73",

```

```

"RDIF" = "#7BAFD4",
"True" = "grey50")

delta_method_map <- c(
  "LRT-unit" = "LRT",
  "RDIF-BH-start" = "RDIF",
  "Oracle-unit" = "True")

item_method_map <- c(
  "LRT" = "LRT",
  "RDIF-BH-start" = "RDIF")

theme_set(theme_bw(base_size = 14))

```

Run Sims

The production run was carried out condition-by-condition. Each condition was saved separately as an `.rds`.

```

SIM1_REPS=500 SIM1_CORES=10 SIM1_LRT_TOL=1e-3 \
Rscript run_sim1.R

```

Simulation 1 uses 500 replications in each cell of a 3 x 11 grid:

- sample size per group: 200, 350, 500;
- number of DIF items: 0, 1, ..., 10;
- total number of items: 20;
- per-item DIF magnitude: 0.5 on the difficulty scale used in `data_sim1()`;
- reference latent mean and variance fixed at 0 and 1;
- comparison latent mean sampled from $U(-0.25, 0.25)$;
- comparison latent variance sampled from $U(0.5, 1.5)$.

The LRT screening step uses joint slope/intercept item tests with `TOL = 1e-3`; this was done to improve run time and piloting showed it had a negligible effect on convergence. The separate single-group `mirt(..., SE = TRUE)` fits used by RDIF are left at the `mirt` defaults.

Load Results

```

sim_files <- list.files(
  results_dir,
  pattern = sim1_pattern,
  full.names = TRUE)

sim_list <- lapply(sim_files, readRDS)
sim_delta_all <- bind_rows(lapply(sim_list, `[`, "delta")) |>
  mutate(
    raw_method = as.character(method),
    n_persons = n_persons_1,
    n_dif = as.integer(n_dif))

sim_delta <- sim_delta_all |>
  filter(raw_method %in% names(delta_method_map)) |>
  mutate(
    method = factor(unname(delta_method_map[raw_method]), levels = method_levels))

sim_items_all <- bind_rows(lapply(sim_list, `[`, "items")) |>
  mutate(
    raw_method = as.character(method),
    n_persons = n_persons_1,
    n_dif = as.integer(n_dif))

sim_items <- sim_items_all |>
  filter(raw_method %in% names(item_method_map)) |>
  mutate(
    method = factor(unname(item_method_map[raw_method]),
      levels = c("LRT", "RDIF")))

run_info <- bind_rows(lapply(sim_list, function(x) {
  data.frame(
    n.reps = x$run$n.reps,
    n.cores = x$run$n.cores,
    base.seed = x$run$base.seed,
    condition.seed = x$run$condition.seed,
    lrt.TOL = x$run$lrt.TOL,
    save.mle = x$run$save.mle,
    completed = x$run$completed)
}))

status_file <- file.path(results_dir, "sim1_status.csv")

```

```
sim_status <- if (file.exists(status_file)) read.csv(status_file) else NULL

n_reps <- length(unique(sim_delta$rep))
n_items <- unique(sim_delta$n_items)
alpha <- 0.05
```

Item-Level Diagnostics

```
item_diag <- sim_items |>
  summarize(
    flagged_rate = mean(flagged, na.rm = TRUE),
    mean_p = mean(p_value, na.rm = TRUE),
    mean_p_adjusted = mean(p_adjusted, na.rm = TRUE),
    .by = c(method, n_persons, n_dif, true_dif)) |>
  mutate(
    diagnostic = ifelse(true_dif, "True positive rate", "False positive rate"))

knitr::kable(item_diag, digits = 3)
```

method	n_persons	n_dif	true_dif	flagged_rate	mean_p	mean_p_adjusted	diagnostic
LRT	200	0	FALSE	0.003	0.495	0.798	False positive rate
RDIF	200	0	FALSE	0.003	0.506	0.815	False positive rate
LRT	200	1	FALSE	0.002	0.498	0.780	False positive rate
LRT	200	1	TRUE	0.200	0.142	0.417	True positive rate
RDIF	200	1	FALSE	0.004	0.505	0.792	False positive rate
RDIF	200	1	TRUE	0.250	0.151	0.399	True positive rate
LRT	200	2	FALSE	0.004	0.496	0.757	False positive rate
LRT	200	2	TRUE	0.201	0.162	0.408	True positive rate
RDIF	200	2	FALSE	0.007	0.506	0.771	False positive rate

method	n_persons	n_dif	true_dif	flagged_rate	mean_p	mean_p_adjusted	diagnostic
RDIF	200	2	TRUE	0.225	0.161	0.410	True positive rate
LRT	200	3	TRUE	0.216	0.163	0.385	True positive rate
LRT	200	3	FALSE	0.005	0.493	0.733	False positive rate
RDIF	200	3	TRUE	0.243	0.158	0.369	True positive rate
RDIF	200	3	FALSE	0.006	0.502	0.748	False positive rate
LRT	200	4	FALSE	0.006	0.492	0.719	False positive rate
LRT	200	4	TRUE	0.187	0.177	0.387	True positive rate
RDIF	200	4	FALSE	0.009	0.508	0.741	False positive rate
RDIF	200	4	TRUE	0.248	0.178	0.376	True positive rate
LRT	200	5	FALSE	0.008	0.473	0.679	False positive rate
LRT	200	5	TRUE	0.146	0.204	0.405	True positive rate
RDIF	200	5	FALSE	0.013	0.502	0.720	False positive rate
RDIF	200	5	TRUE	0.238	0.203	0.392	True positive rate
LRT	200	6	FALSE	0.009	0.465	0.656	False positive rate
LRT	200	6	TRUE	0.140	0.212	0.398	True positive rate
RDIF	200	6	FALSE	0.021	0.492	0.691	False positive rate
RDIF	200	6	TRUE	0.240	0.209	0.377	True positive rate
LRT	200	7	FALSE	0.014	0.440	0.621	False positive rate
LRT	200	7	TRUE	0.106	0.248	0.432	True positive rate
RDIF	200	7	FALSE	0.024	0.483	0.673	False positive rate

method	n_persons	n_dif	true_dif	flagged_rate	mean_p	mean_p_adjusted	diagnostic
RDIF	200	7	TRUE	0.219	0.236	0.399	True positive rate
LRT	200	8	TRUE	0.076	0.275	0.453	True positive rate
LRT	200	8	FALSE	0.019	0.414	0.585	False positive rate
RDIF	200	8	TRUE	0.202	0.265	0.423	True positive rate
RDIF	200	8	FALSE	0.043	0.464	0.640	False positive rate
LRT	200	9	FALSE	0.023	0.383	0.550	False positive rate
LRT	200	9	TRUE	0.065	0.310	0.479	True positive rate
RDIF	200	9	FALSE	0.060	0.439	0.609	False positive rate
RDIF	200	9	TRUE	0.187	0.301	0.455	True positive rate
LRT	200	10	TRUE	0.033	0.358	0.526	True positive rate
LRT	200	10	FALSE	0.044	0.342	0.509	False positive rate
RDIF	200	10	TRUE	0.126	0.361	0.523	True positive rate
RDIF	200	10	FALSE	0.100	0.387	0.553	False positive rate
LRT	350	0	FALSE	0.003	0.497	0.803	False positive rate
RDIF	350	0	FALSE	0.005	0.509	0.817	False positive rate
LRT	350	1	FALSE	0.004	0.503	0.782	False positive rate
LRT	350	1	TRUE	0.462	0.059	0.237	True positive rate
RDIF	350	1	FALSE	0.005	0.506	0.787	False positive rate
RDIF	350	1	TRUE	0.472	0.071	0.245	True positive rate
LRT	350	2	FALSE	0.004	0.501	0.757	False positive rate

method	n_persons	n_dif	true_dif	flagged_rate	mean_p	mean_p_adjusted	diagnostic
LRT	350	2	TRUE	0.435	0.079	0.235	True positive rate
RDIF	350	2	FALSE	0.008	0.511	0.769	False positive rate
RDIF	350	2	TRUE	0.464	0.079	0.242	True positive rate
LRT	350	3	TRUE	0.466	0.078	0.210	True positive rate
LRT	350	3	FALSE	0.007	0.498	0.728	False positive rate
RDIF	350	3	TRUE	0.486	0.079	0.212	True positive rate
RDIF	350	3	FALSE	0.010	0.512	0.744	False positive rate
LRT	350	4	FALSE	0.009	0.491	0.698	False positive rate
LRT	350	4	TRUE	0.441	0.093	0.214	True positive rate
RDIF	350	4	FALSE	0.013	0.507	0.719	False positive rate
RDIF	350	4	TRUE	0.509	0.089	0.201	True positive rate
LRT	350	5	FALSE	0.012	0.473	0.653	False positive rate
LRT	350	5	TRUE	0.420	0.099	0.211	True positive rate
RDIF	350	5	FALSE	0.022	0.506	0.696	False positive rate
RDIF	350	5	TRUE	0.526	0.090	0.190	True positive rate
LRT	350	6	TRUE	0.348	0.117	0.231	True positive rate
LRT	350	6	FALSE	0.013	0.462	0.626	False positive rate
RDIF	350	6	TRUE	0.518	0.096	0.188	True positive rate
RDIF	350	6	FALSE	0.019	0.509	0.684	False positive rate
LRT	350	7	FALSE	0.024	0.426	0.567	False positive rate

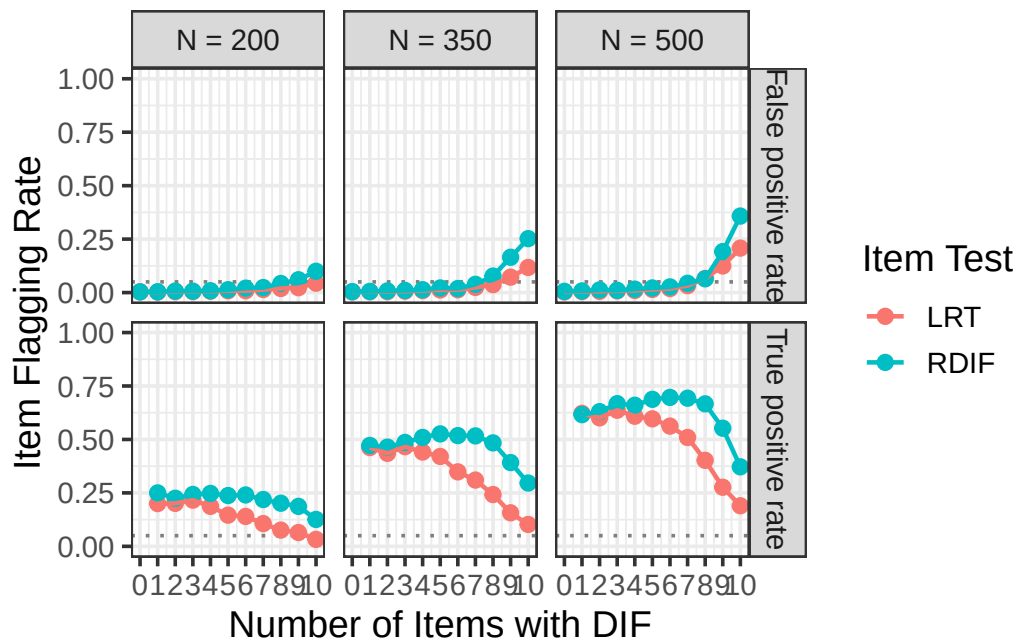
method	n_persons	n_dif	true_dif	flagged_rate	mean_p	mean_p_adjusted	diagnostic
LRT	350	7	TRUE	0.310	0.145	0.253	True positive rate
RDIF	350	7	FALSE	0.038	0.499	0.657	False positive rate
RDIF	350	7	TRUE	0.517	0.114	0.197	True positive rate
LRT	350	8	TRUE	0.242	0.179	0.284	True positive rate
LRT	350	8	FALSE	0.037	0.385	0.510	False positive rate
RDIF	350	8	TRUE	0.484	0.158	0.240	True positive rate
RDIF	350	8	FALSE	0.077	0.477	0.619	False positive rate
LRT	350	9	FALSE	0.072	0.320	0.429	False positive rate
LRT	350	9	TRUE	0.157	0.228	0.330	True positive rate
RDIF	350	9	FALSE	0.165	0.418	0.540	False positive rate
RDIF	350	9	TRUE	0.392	0.239	0.328	True positive rate
LRT	350	10	TRUE	0.102	0.284	0.390	True positive rate
LRT	350	10	FALSE	0.117	0.268	0.375	False positive rate
RDIF	350	10	TRUE	0.296	0.317	0.419	True positive rate
RDIF	350	10	FALSE	0.252	0.345	0.451	False positive rate
LRT	500	0	FALSE	0.003	0.500	0.804	False positive rate
RDIF	500	0	FALSE	0.006	0.505	0.810	False positive rate
LRT	500	1	FALSE	0.005	0.503	0.781	False positive rate
LRT	500	1	TRUE	0.622	0.041	0.157	True positive rate
RDIF	500	1	FALSE	0.008	0.507	0.784	False positive rate

method	n_persons	n_dif	true_dif	flagged_rate	mean_p	mean_p_adjusted	diagnostic
RDIF	500	1	TRUE	0.616	0.039	0.158	True positive rate
LRT	500	2	FALSE	0.006	0.498	0.745	False positive rate
LRT	500	2	TRUE	0.601	0.042	0.147	True positive rate
RDIF	500	2	FALSE	0.012	0.502	0.750	False positive rate
RDIF	500	2	TRUE	0.630	0.041	0.140	True positive rate
LRT	500	3	FALSE	0.008	0.497	0.723	False positive rate
LRT	500	3	TRUE	0.637	0.049	0.137	True positive rate
RDIF	500	3	FALSE	0.011	0.508	0.735	False positive rate
RDIF	500	3	TRUE	0.667	0.046	0.124	True positive rate
LRT	500	4	FALSE	0.010	0.496	0.699	False positive rate
LRT	500	4	TRUE	0.609	0.055	0.134	True positive rate
RDIF	500	4	FALSE	0.016	0.512	0.719	False positive rate
RDIF	500	4	TRUE	0.660	0.047	0.119	True positive rate
LRT	500	5	FALSE	0.015	0.478	0.655	False positive rate
LRT	500	5	TRUE	0.596	0.059	0.132	True positive rate
RDIF	500	5	FALSE	0.022	0.511	0.697	False positive rate
RDIF	500	5	TRUE	0.688	0.049	0.108	True positive rate
LRT	500	6	FALSE	0.019	0.458	0.609	False positive rate
LRT	500	6	TRUE	0.562	0.071	0.141	True positive rate
RDIF	500	6	FALSE	0.027	0.511	0.680	False positive rate

method	n_persons	n_dif	true_dif	flagged_rate	mean_p	mean_p_adjusted	diagnostic
RDIF	500	6	TRUE	0.697	0.055	0.109	True positive rate
LRT	500	7	FALSE	0.033	0.425	0.550	False positive rate
LRT	500	7	TRUE	0.509	0.090	0.156	True positive rate
RDIF	500	7	FALSE	0.043	0.504	0.652	False positive rate
RDIF	500	7	TRUE	0.693	0.068	0.115	True positive rate
LRT	500	8	TRUE	0.402	0.124	0.192	True positive rate
LRT	500	8	FALSE	0.066	0.366	0.468	False positive rate
RDIF	500	8	TRUE	0.666	0.091	0.139	True positive rate
RDIF	500	8	FALSE	0.064	0.502	0.636	False positive rate
LRT	500	9	FALSE	0.124	0.288	0.370	False positive rate
LRT	500	9	TRUE	0.276	0.174	0.245	True positive rate
RDIF	500	9	FALSE	0.192	0.423	0.532	False positive rate
RDIF	500	9	TRUE	0.552	0.189	0.247	True positive rate
LRT	500	10	FALSE	0.208	0.208	0.279	False positive rate
LRT	500	10	TRUE	0.190	0.231	0.301	True positive rate
RDIF	500	10	FALSE	0.358	0.316	0.396	False positive rate
RDIF	500	10	TRUE	0.372	0.307	0.388	True positive rate

```
ggplot(
  item_diag,
  aes(x = n_dif, y = flagged_rate, color = method, group = method)) +
  geom_hline(yintercept = alpha, linetype = 3, color = "grey50") +
  geom_point(size = 2.2) +
```

```
geom_line(linewidth = 0.8) +
scale_x_continuous(breaks = 0:10) +
scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, 0.25)) +
facet_grid(
  diagnostic ~ n_persons,
  labeller = labeller(n_persons = function(x) paste0("N = ", x))) +
labs(
  x = "Number of Items with DIF",
  y = "Item Flagging Rate",
  color = "Item Test")
```



Bias and Precision

Scaling Function Bias

Figure 2 from the paper. Naive is the mean of all item-level scaling functions. The other methods use the method-specific estimate of the common-item scaling function.

```
scale_bias <- bind_rows(
  sim_delta |>
  filter(method == "RDIF") |>
  transmute(
```

```

    Method = "Naive",
    n_persons,
    n_dif,
    Bias = naive_est - delta_0),
sim_delta |>
  transmute(
    Method = as.character(method),
    n_persons,
    n_dif,
    Bias = scale_est - delta_0)) |>
mutate(
  Method = factor(
    Method,
    levels = c("Naive", method_levels),
    ordered = TRUE))

scale_cols <- c("Naive" = "grey80", method_cols)

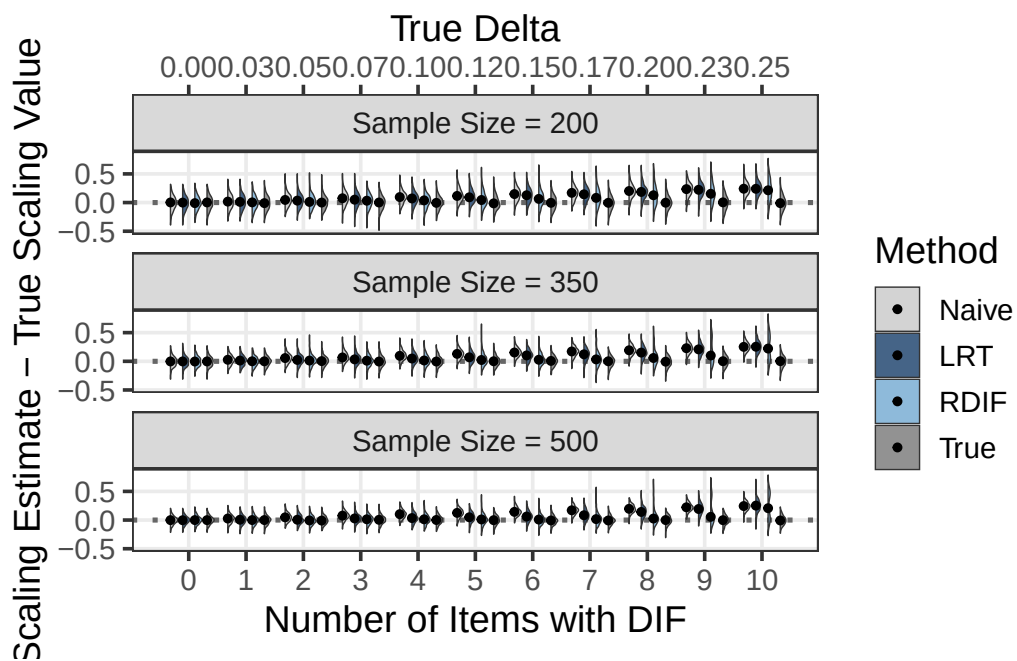
ggplot(scale_bias, aes(x = n_dif, y = Bias, fill = Method)) +
  geom_hline(yintercept = 0, linetype = 3, linewidth = 0.9, color = "grey40") +
  see::geom_violinhalf(
    aes(group = interaction(n_dif, Method)),
    position = position_dodge(width = 0.85),
    trim = TRUE,
    width = 2,
    linewidth = 0.25,
    alpha = 0.85) +
  stat_summary(
    aes(group = interaction(n_dif, Method)),
    fun = median,
    geom = "point",
    position = position_dodge(width = 0.85),
    size = 0.9,
    color = "black") +
  scale_x_continuous(
    breaks = 0:10,
    sec.axis = sec_axis(
      ~ . * 0.5 / 20,
      breaks = (0:10) * 0.5 / 20,
      labels = sprintf("%.2f", (0:10) * 0.5 / 20),
      name = "True Delta")) +
  scale_fill_manual(values = scale_cols) +

```

```

facet_wrap(
  ~ n_persons,
  nrow = 3,
  scales = "free_y",
  labeller = labeller(n_persons = function(x) paste0("Sample Size = ", x))) +
labs(
  x = "Number of Items with DIF",
  y = "Scaling Estimate - True Scaling Value",
  fill = "Method") +
theme(panel.grid.minor = element_blank())

```



```

scale_bias_table <- scale_bias |>
  summarize(
    bias = mean(Bias, na.rm = TRUE),
    .by = c(Method, n_persons, n_dif)) |>
  mutate(Method = as.character(Method)) |>
  arrange(n_persons, n_dif, Method) |>
  pivot_wider(names_from = Method, values_from = bias)

knitr::kable(scale_bias_table, digits = 3)

```

n_persons	n_dif	LRT	Naive	RDIF	True
200	0	-0.007	-0.006	-0.007	-0.006
200	1	0.013	0.021	0.001	-0.005
200	2	0.035	0.049	0.016	-0.001
200	3	0.050	0.071	0.027	-0.005
200	4	0.072	0.097	0.034	-0.003
200	5	0.100	0.122	0.054	-0.001
200	6	0.123	0.147	0.071	-0.002
200	7	0.147	0.167	0.089	-0.007
200	8	0.189	0.202	0.137	0.000
200	9	0.219	0.230	0.171	0.001
200	10	0.245	0.242	0.223	-0.005
350	0	-0.002	-0.003	-0.001	-0.003
350	1	0.010	0.024	0.002	-0.002
350	2	0.027	0.053	0.012	0.003
350	3	0.036	0.076	0.014	0.001
350	4	0.051	0.101	0.017	0.000
350	5	0.072	0.129	0.030	0.003
350	6	0.099	0.155	0.037	0.005
350	7	0.117	0.171	0.042	-0.005
350	8	0.150	0.193	0.077	-0.009
350	9	0.209	0.228	0.160	0.002
350	10	0.255	0.252	0.238	-0.002
500	0	-0.002	-0.002	-0.001	-0.002
500	1	0.006	0.023	-0.001	-0.002
500	2	0.013	0.046	-0.003	-0.004
500	3	0.028	0.079	0.010	0.005
500	4	0.038	0.102	0.014	0.003
500	5	0.046	0.123	0.011	-0.001
500	6	0.063	0.148	0.016	-0.001
500	7	0.087	0.171	0.025	-0.004
500	8	0.137	0.201	0.049	0.002
500	9	0.193	0.225	0.132	0.000
500	10	0.251	0.248	0.240	-0.005

DTF Effect Bias

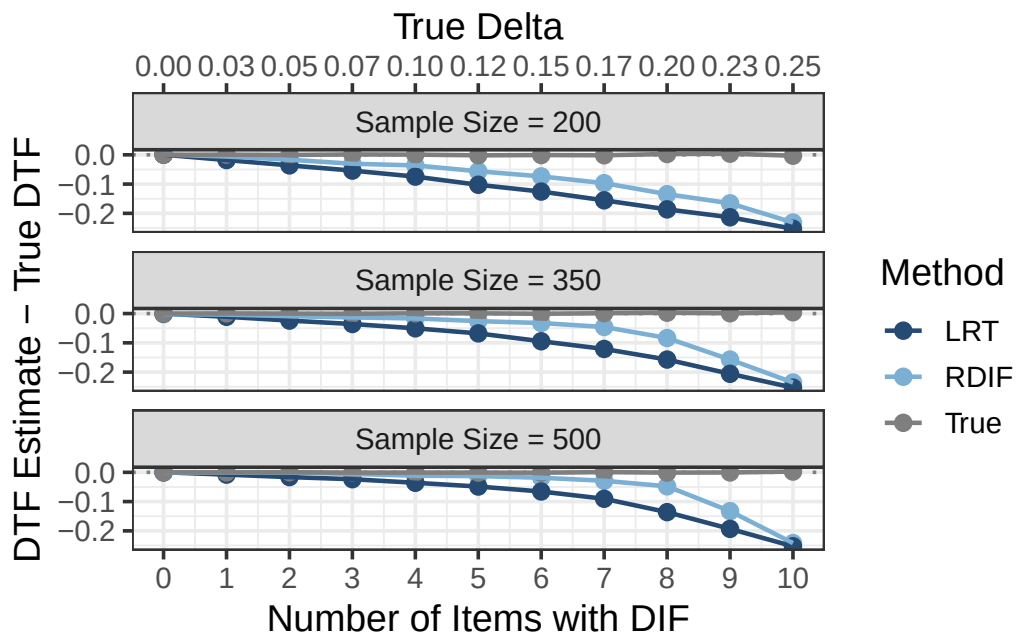
```
delta_bias <- sim_delta |>
  mutate(Bias = delta_hat - true_Delta) |>
  summarize(
```

```

bias = mean(Bias, na.rm = TRUE),
.by = c(method, n_persons, n_dif))

ggplot(delta_bias, aes(x = n_dif, y = bias, color = method, group = method)) +
  geom_hline(yintercept = 0, linetype = 3, color = "grey50") +
  geom_point(size = 2.4) +
  geom_line(linewidth = 0.8) +
  scale_x_continuous(
    breaks = 0:10,
    sec.axis = sec_axis(
      ~ . * 0.5 / 20,
      breaks = (0:10) * 0.5 / 20,
      labels = sprintf("%.2f", (0:10) * 0.5 / 20),
      name = "True Delta")) +
  scale_color_manual(values = method_cols) +
  facet_wrap(
    ~ n_persons,
    nrow = 3,
    labeller = labeller(n_persons = function(x) paste0("Sample Size = ", x))) +
  labs(
    x = "Number of Items with DIF",
    y = "DTF Estimate - True DTF",
    color = "Method")

```



```

delta_bias_wide <- delta_bias |>
  mutate(method = as.character(method)) |>
  arrange(n_persons, n_dif, method) |>
  select(method, n_persons, n_dif, bias) |>
  pivot_wider(names_from = method, values_from = bias)

knitr::kable(delta_bias_wide, digits = 3)

```

n_persons	n_dif	LRT	RDIF	True
200	0	0.000	0.001	0.000
200	1	-0.018	-0.006	0.001
200	2	-0.036	-0.017	0.000
200	3	-0.054	-0.031	0.001
200	4	-0.075	-0.036	0.001
200	5	-0.102	-0.056	-0.002
200	6	-0.126	-0.074	-0.001
200	7	-0.155	-0.097	-0.002
200	8	-0.186	-0.134	0.003
200	9	-0.213	-0.166	0.004
200	10	-0.253	-0.231	-0.003
350	0	0.000	-0.002	0.000
350	1	-0.011	-0.004	0.001
350	2	-0.024	-0.009	0.000
350	3	-0.036	-0.013	-0.001
350	4	-0.050	-0.017	0.001
350	5	-0.068	-0.026	0.001
350	6	-0.095	-0.032	-0.001
350	7	-0.121	-0.046	0.001
350	8	-0.157	-0.084	0.003
350	9	-0.206	-0.157	0.001
350	10	-0.253	-0.236	0.004
500	0	0.000	-0.001	0.000
500	1	-0.008	-0.001	0.000
500	2	-0.016	-0.001	0.000
500	3	-0.024	-0.006	-0.001
500	4	-0.036	-0.012	-0.001
500	5	-0.049	-0.013	-0.001
500	6	-0.066	-0.018	-0.001
500	7	-0.090	-0.029	0.001
500	8	-0.136	-0.048	-0.001
500	9	-0.193	-0.133	0.000

n_persons	n_dif	LRT	RDIF	True
500	10	-0.253	-0.242	0.003

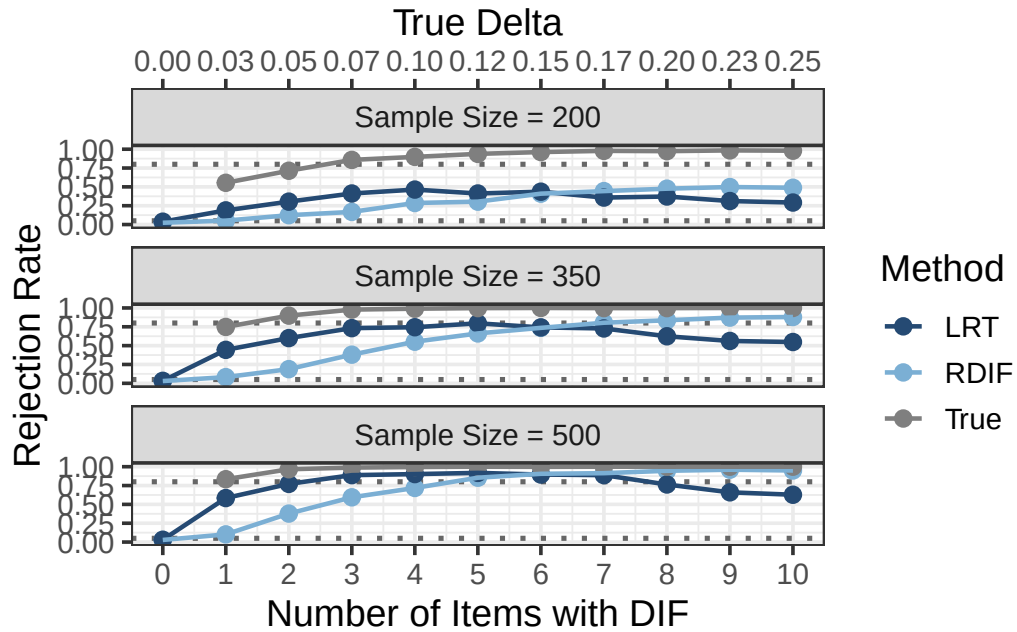
Rejection Rates

```

sim_power <- sim_delta |>
  summarize(
    rejection_rate = mean(rejected, na.rm = TRUE),
    .by = c(method, n_persons, n_dif)) |>
  mutate(
    rejection_rate = ifelse(
      method == "True" & n_dif == 0,
      NA_real_,
      rejection_rate))

ggplot(sim_power, aes(x = n_dif, y = rejection_rate, color = method, group = method)) +
  geom_hline(yintercept = 0.05, linetype = 3, linewidth = 0.9, color = "grey40") +
  geom_hline(yintercept = 0.80, linetype = 3, linewidth = 0.9, color = "grey40") +
  geom_point(size = 2.4, na.rm = TRUE) +
  geom_line(linewidth = 0.8, na.rm = TRUE) +
  scale_x_continuous(
    breaks = 0:10,
    sec.axis = sec_axis(
      ~ . * 0.5 / 20,
      breaks = (0:10) * 0.5 / 20,
      labels = sprintf("%.2f", (0:10) * 0.5 / 20),
      name = "True Delta")) +
  scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, .25)) +
  scale_color_manual(values = method_cols) +
  facet_wrap(
    ~ n_persons,
    nrow = 3,
    labeller = labeller(n_persons = function(x) paste0("Sample Size = ", x))) +
  labs(
    x = "Number of Items with DIF",
    y = "Rejection Rate",
    color = "Method")

```



```
sim_power_wide <- sim_power |>
  pivot_wider(names_from = method, values_from = rejection_rate)

knitr::kable(sim_power_wide, digits = 3)
```

n_persons	n_dif	RDIF	LRT	True
200	0	0.024	0.038	NA
200	1	0.052	0.188	0.554
200	2	0.124	0.304	0.714
200	3	0.168	0.412	0.858
200	4	0.286	0.464	0.898
200	5	0.304	0.412	0.938
200	6	0.410	0.436	0.962
200	7	0.444	0.358	0.978
200	8	0.476	0.372	0.974
200	9	0.498	0.312	0.984
200	10	0.488	0.292	0.980
350	0	0.030	0.032	NA
350	1	0.082	0.444	0.748
350	2	0.188	0.600	0.898
350	3	0.380	0.732	0.978
350	4	0.552	0.744	0.990
350	5	0.662	0.796	0.998

n_persons	n_dif	RDIF	LRT	True
350	6	0.736	0.742	1.000
350	7	0.804	0.728	0.998
350	8	0.836	0.624	0.998
350	9	0.872	0.562	1.000
350	10	0.880	0.548	1.000
500	0	0.028	0.030	NA
500	1	0.102	0.584	0.834
500	2	0.378	0.772	0.966
500	3	0.594	0.888	0.988
500	4	0.718	0.902	0.996
500	5	0.854	0.918	1.000
500	6	0.906	0.896	0.998
500	7	0.914	0.888	1.000
500	8	0.946	0.764	1.000
500	9	0.960	0.660	1.000
500	10	0.950	0.628	1.000

Coverage

```

sim_coverage <- sim_delta |>
  summarize(
    coverage = mean(covered, na.rm = TRUE),
    mean_width = mean(2 * qnorm(.975) * delta_se, na.rm = TRUE),
    mean_se = mean(delta_se, na.rm = TRUE),
    .by = c(method, n_persons, n_dif))

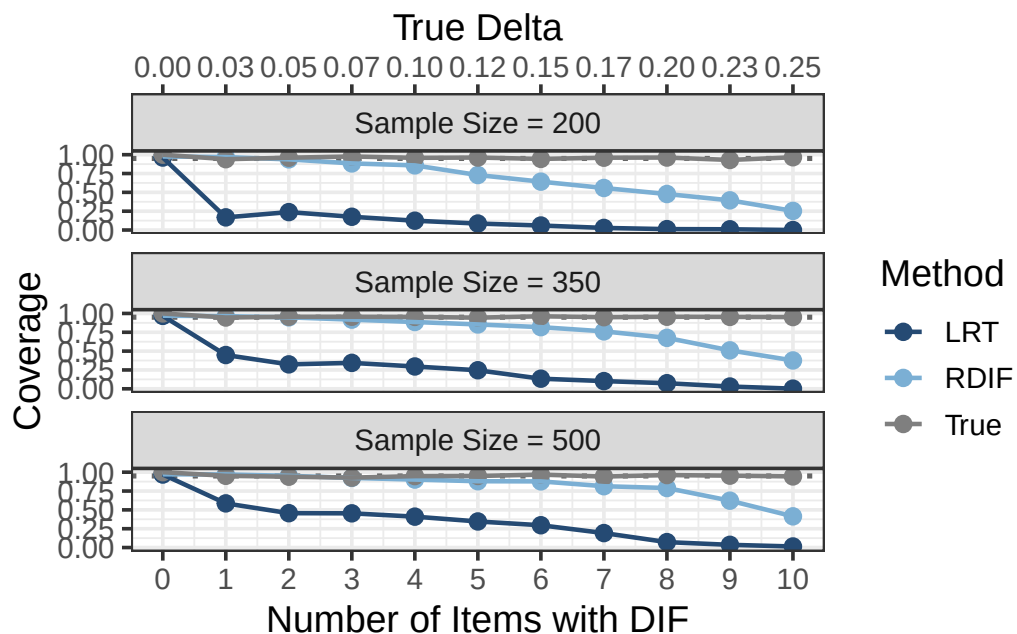
ggplot(sim_coverage, aes(x = n_dif, y = coverage, color = method, group = method)) +
  geom_hline(yintercept = 0.95, linetype = 3, linewidth = 0.9, color = "grey40") +
  geom_point(size = 2.4) +
  geom_line(linewidth = 0.8) +
  scale_x_continuous(
    breaks = 0:10,
    sec.axis = sec_axis(
      ~ . * 0.5 / 20,
      breaks = (0:10) * 0.5 / 20,
      labels = sprintf("%.2f", (0:10) * 0.5 / 20),
      name = "True Delta")) +
  scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, .25)) +

```

```

scale_color_manual(values = method_cols) +
facet_wrap(
  ~ n_persons,
  nrow = 3,
  labeller = labeller(n_persons = function(x) paste0("Sample Size = ", x))) +
labs(
  x = "Number of Items with DIF",
  y = "Coverage",
  color = "Method")

```



```

sim_coverage_wide <- sim_coverage |>
  select(method, n_persons, n_dif, coverage) |>
  pivot_wider(names_from = method, values_from = coverage)

knitr::kable(sim_coverage_wide, digits = 3)

```

n_persons	n_dif	RDIF	LRT	True
200	0	0.976	0.962	1.000
200	1	0.968	0.168	0.940
200	2	0.938	0.238	0.960
200	3	0.884	0.176	0.976
200	4	0.858	0.124	0.958

n_persons	n_dif	RDIF	LRT	True
200	5	0.730	0.086	0.962
200	6	0.642	0.060	0.944
200	7	0.558	0.028	0.960
200	8	0.478	0.012	0.962
200	9	0.394	0.010	0.930
200	10	0.254	0.000	0.966
350	0	0.970	0.968	1.000
350	1	0.964	0.448	0.946
350	2	0.948	0.324	0.956
350	3	0.918	0.344	0.956
350	4	0.886	0.296	0.954
350	5	0.854	0.246	0.944
350	6	0.818	0.134	0.962
350	7	0.762	0.102	0.950
350	8	0.676	0.074	0.956
350	9	0.510	0.030	0.954
350	10	0.376	0.002	0.952
500	0	0.972	0.970	1.000
500	1	0.970	0.586	0.952
500	2	0.952	0.456	0.940
500	3	0.924	0.454	0.928
500	4	0.902	0.410	0.948
500	5	0.880	0.346	0.950
500	6	0.878	0.296	0.968
500	7	0.814	0.192	0.942
500	8	0.790	0.072	0.962
500	9	0.622	0.036	0.954
500	10	0.414	0.014	0.944